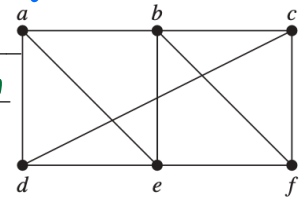
MAT2540, Classwork17, Spring2026

10.4 Connectivity

1. Introduction of Path:

   A path is a sequence of edges that begins at a vertex of a graph and travels from vertex to vertex along edges of graph. As the path travels along its edges, it visits the vertices along this path, that is, the endpoints of these edges.

2. Example of paths.



   (1) a, b, c, d, e, f is a path of length 5 since $\{a,b\}$, $\{b,c\}$, $\{c,d\}$, $\{d,e\}$, $\{e,f\}$ are edges in this graph

   (2) d, e, c, a is not a path since $\{e,c\}$ is not an edge in this graph

   (3) b, c, f, e, b is a path of length 4 since $\{b,c\}$, $\{c,f\}$, $\{f,e\}$, $\{e,b\}$ are edges

   (4) a, b, e, d, a, b is a path of length 5 and it goes through $\{a,b\}$ twice.

3. *Definition:* The **Path** of an **Undirected Graph**. Let n be a nonnegative integer and G an undirected graph.

   A *path of length* n from u to v in G is a sequence of n edges $e_1, e_2, \dots, e_n$ for which there exists a sequence of vertices u $x_0, x_1, x_2, \dots, x_n$ v such that $e_i = \{x_{i-1}, x_i\}$.

   When the graph is *simple*, we denote this path by its vertices $x_0, x_1, x_2, \dots, x_n$ because listing these vertices uniquely determines the path

   The path is *a circuit* if begins and ends at the vertex, that is, if $u = v$, and has length greater than zero.

   The path or circuit is said to pass through the vertices $x_1, x_2, \dots, x_n$ or traverse the edges $e_1, e_2, \dots, e_n$.

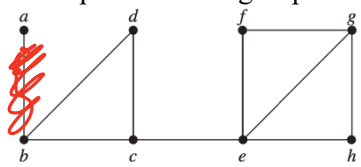
   A path or circuit is *simple* if it does not contain the same edge more than once

4. *Definition:* The **Subgraph** of A Graph

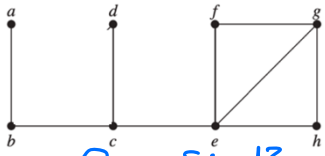
   A subgraph of a graph $G = (V, E)$ is a graph $H = (W, F)$, where $W \subseteq V$ and $F \subseteq E$.

   A subgraph H of G is a proper subgraph of G if $H \neq G$.

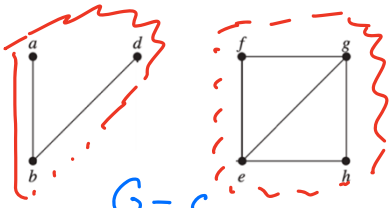
5. Example of the subgroup of a graph



G



$G - \{b,d\}$



$G - c$

6. *Definition:* The **Connected Graph**

   An undirected graph is called *connected* if there is a path between every pair of distinct vertices of the graph.

   An undirected graph that is not connected is called disconnected. We say that we disconnect a graph when we remove vertices or edges, or both, to produce a disconnected subgraph

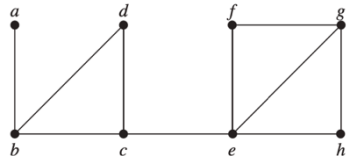
7. *Theorem:* There is a simple path between every pair of distinct vertices of a connected undirected graph.

8. The **Connected Component of a Graph**

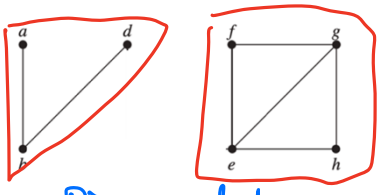
   If a graph G is not connected, then it has 2 or more connected components that are disjoint and have G as their union.

   Each connected component of G is not a proper subgraph of another connected component of G.

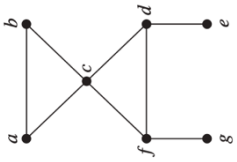
9. Examples. Check if each given graph is connected. If it is not connected, find its connected components.



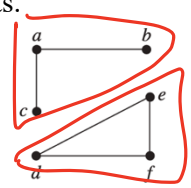
Connected



Disconnected with 2 connected components



Connected



Disconnected with 2 connected components

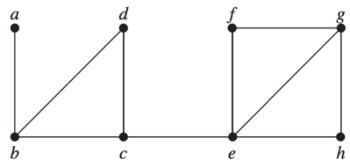
10. **Cut Vertex** and **Cut Edge**.

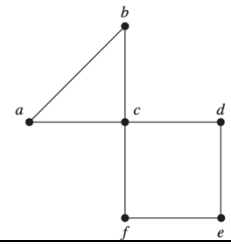
and all incident edges

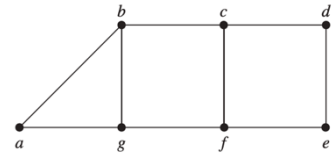
Cut Vertex: A vertex is called *cut vertex* if the removal from a graph of that vertex produces a subgraph with more connected components. The removal of a cut vertex from a connected graph produces a subgraph that is connected.

Cut edge: An edge is called a *cut edge* or bridge if the edge removal produces a graph with more connected components than in the original graph. If a graph has a cut edge, then we need only remove it to disconnect G.

11. Find the cut vertex and cut edge of each given graph.







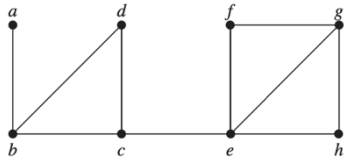
Cut Vertex	b or c or e	c	no cut vertex
Cut Edge	{c,e} or {a,b}	no cut edge	no cut edge

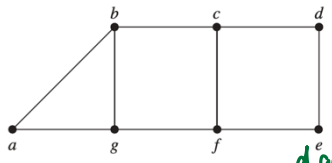
12. **Vertex Cut** and **Vertex Connectivity**.

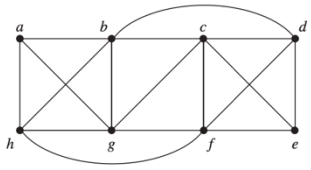
kappa κ

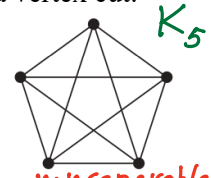
Vertex cut: A subset V' of the vertex set V of $G = (V, E)$ is a vertex cut, or separating set, if $G - V'$ is disconnected.

Vertex connectivity of a noncomplete graph G, denoted by $\kappa(G)$, is the minimum number of vertices in a vertex cut.









Vertex cut	c or e or b	f and c, b and g, d and f	b and g and f or b, c, and f	nonseparable no vertex cut
Vertex connectivity	$\kappa(G) = 1$	2	3	$5-1=4$

To sum up, $\kappa(G)$ is minimum number of vertices that can be removed from G to either disconnect G or produce a graph with a single vertex. We have the larger $\kappa(G)$ is, the more connected we consider G to be

(1) $0 \le \kappa(G) \le n-1$ if G has n vertices, (2) $\kappa(G)=0$ if and only if G is disconnected or $G=K_1$ (single vertex), and (3) $\kappa(G)=n-1$ if and only if G is complete with n vertices.

13. **Edge Cut** and **Edge Connectivity**.

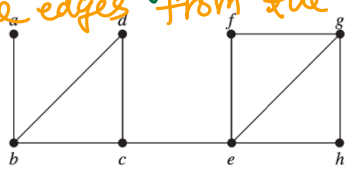
lambda (λ)

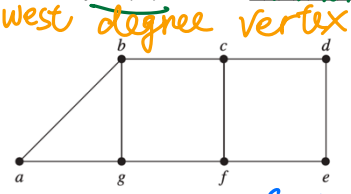
If G does not have a cut edge, we look for the smallest set of edges that can be removed to disconnect it.

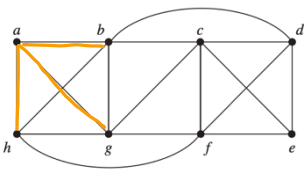
Edge Cut: A set of edges E' is called an edge cut of G if the subgraph $G - E'$ is disconnected.

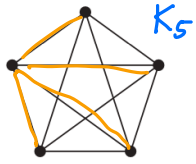
Edge connectivity of a graph G, denoted by $\lambda(G)$, is the minimum number of edges in an edge cut of G.

• cut the edges from the lowest degree vertex









Edge cut	{c,e} or {a,b}	{b,c} and {g,f}, {c,d} and {f,e}	{a,h} and {a,g} and {a,b}	
Edge connectivity	1	2	3	4

To sum up, (1) $0 \le \lambda(G) \le n-1$ if G has n vertices, (2) $\lambda(G)=0$ if and only if G is disconnected or $G=K_1$ (single point vertex), (3) $\lambda(G)=n-1$ if and only if G is complete with n vertices.

14. An Inequality For Vertex Connectivity and Edge Connectivity: For a graph $G = (V, E)$, we have

$$\kappa(G) \le \lambda(G) \le \min_{v \in V} \deg(v).$$